

Find the general solution of each differential equation.

$$y \frac{dy}{dx} =$$

$$y \frac{dy}{dx} = \frac{1}{\sec^2 x}$$

$$\frac{dy}{dx} = 3e^x$$

$$y \frac{dy}{dx} = \frac{1}{x}$$

$$= \frac{2x}{v^2}, y(2) =$$

$$2e^{x-y}, y(-3) = 1$$

$$= \frac{1}{\sec^2 v}, y(3)$$

$$\frac{e^x}{v^2}, y(-1) = \frac{1}{2}$$

$$y' = -\frac{1}{\sin v}, y(3)$$

$$\frac{dy}{dx} = \frac{2x}{e^{2y}}, y(2)$$

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$$= \frac{1+x^2}{v^2}, y(-1) =$$

$$\begin{aligned}
 & C_1 y \\
 & + \frac{1}{3} y = \ln(2e^x + 1) \\
 & \frac{1}{3} y = \ln(2e^x + 1) - C_1 y \\
 & \frac{1}{3} y + C_1 y = \ln(2e^x + 1) \\
 & y \left(\frac{1}{3} + C_1 \right) = \ln(2e^x + 1) \\
 & y = \frac{\ln(2e^x + 1)}{\frac{1}{3} + C_1}
 \end{aligned}$$

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$$) \frac{dy}{dx} =$$

$$) \frac{dy}{dx} = \frac{1}{\sec}$$

$$\frac{dy}{dx} = 3e^x$$

$$) \frac{dy}{dx} = \frac{2}{\rho}$$

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any $y = \tan^{-1}(x)$

$$y = \ln(3e^x + C)$$

$$\sqrt{5.4}$$

$$e^x = 2e^x +$$

$$y = \tan^{-1}(x) = \ln(2x^2)$$

$$+ \frac{1}{3} y = \ln(2e^x +$$

$$3 y = \cos^{-1}(x) \frac{e^x}{2} = x^2 - \frac{1}{2}$$

$$\frac{3}{2}$$

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